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DEPARTMENT OF DEFENSE HANDBOOK

NONDESTRUCTIVE INSPECTION PROGRAM REQUIREMENTS FOR AIRCRAFT AND MISSILE MATERIALS AND PARTS



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FOREWORD

1. This handbook is approved for use by all Departments and Agencies of the Department of Defense (DoD).

2. This handbook provides the guidance for defining the requirements for establishing a nondestructive inspection (NDI) program for evaluation of structural components in development, production or sustainment of aircraft and missiles systems. This document is directly applicable when referenced in the item specification contract or order.

3. Comments, suggestions, or questions regarding this document should be addressed to ASC/ENRS, 2145 Monahan Way, Wright-Patterson AFB, OH 45433-7017 or e-mailed to Engineering.Standards@wpafb.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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CONTENTS

Paragraph		Page
1.	**SCOPE**	**1**
1.1	Scope.	1
1.2	Applicability.	1
2.	**APPLICABLE DOCUMENTS**	**1**
2.1	General	1
2.2	Government documents.	1
2.2.1	Specifications, standards, and handbooks.	1
2.2.2	Other Government documents.	2
2.3	Non-Government publications.	2
3.	**DEFINITIONS**	**4**
3.1	Inspector certification.	4
3.2	Classification.	4
3.3	Contracting agency.	4
3.4	Contractor.	4
3.5	Final inspection.	4
3.6	Fracture-critical part.	4
3.7	Fracture-critical traceable part.	4
3.8	Maintenance-critical part.	4
3.9	Mission-critical part.	5
3.10	Material Review Board (MRB).	5
3.11	Nondestructive inspection (NDI).	5
3.12	Nondestructive inspection (NDI) procedure.	5
3.12.1	General NDI procedure.	5
3.12.2	Part specific NDI procedures.	5
3.13	NDI facility.	5
3.14	NDI Program Plan.	5
3.15	NDI process specification.	5
3.16	NDI Requirements Review Board (NDIRRB).	6
3.17	Probability of detection (POD).	6
3.18	Qualification.	6
3.19	Safety-of-flight structure.	6
3.20	Subcontractors.	6
3.21	Supplier.	6
4.	**GENERAL REQUIREMENTS**	**6**
4.1	Preparation of the NDI Program Plan.	7
4.2	Responsibility for NDI.	7
4.2.1	Objective.	7
4.2.2	Applicability.	7
4.2.3	Elements.	7
4.2.4	Coordination.	8
4.3	Parts classification.	8
4.3.1	Structural parts classification.	8
4.3.2	Propulsion systems parts classification.	9
4.3.3	Mechanical equipment and subsystem parts classification.	9
4.4	NDI Requirements Review Board (NDIRRB).	9
4.4.1	Purpose.	9

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Check the source to verify that this is the current version before use.

CONTENTS

Paragraph		Page
4.4.2	NDI Requirements Review Board (NDIRRB) Membership.	9
5.	**DETAILED REQUIREMENTS FOR PRODUCTION**	**9**
5.1	Documentation of the review.	9
5.2	Drawings.	10
5.3	Preparation of NDI procedures and process specifications.	10
5.3.1	Use of general NDI process specifications.	10
5.3.2	Company NDI process specifications.	10
5.3.2.1	Special NDI procedures.	10
5.3.2.2	NDI standardization.	10
5.3.2.3	Approval.	10
5.3.3	NDI processes.	11
5.3.4	NDI procedures.	11
5.3.5	General NDI procedures.	12
5.4	Implementation of NDI procedures.	12
5.4.1	Personnel.	12
5.4.2	NDI reports.	12
5.4.3	Equipment and materials.	12
5.4.4	NDI procedure verification.	12
5.4.5	Removal of discontinuities.	12
5.4.5.1	Reinspection.	12
5.4.6	Inspection stages.	13
5.4.6.1	First article inspections.	13
5.4.6.2	Receiving inspections.	13
5.4.6.3	Manufacturing and assembly.	13
5.4.7	Data and documentation.	13
5.4.8	Vendor qualification.	13
5.4.9	Capability demonstration.	14
6.	**DETAILED REQUIREMENTS FOR TEST ARTICLES**	**14**
6.1	First article testing.	14
6.2	Static test articles.	14
6.3	Component, full-scale durability test articles.	14
6.4	Teardown inspections.	14
6.5	Flight testing.	14
6.6	Test support monitoring inspections.	15
7.	**DETAILED REQUIREMENTS FOR SUSTAINMENT**	**15**
7.1	Defining sustainment inspection requirements.	15
7.2	Sustainment inspection requirements review.	15
7.3	Preparation of NDI procedures.	15
7.3.1	Sustainment NDI procedure content.	15
7.3.2	Sustainment NDI procedure format.	16
7.4	Inspection capability demonstration.	16
7.5	NDI procedure validation and verification.	16
7.5.1	Procedure validation.	17
7.5.2	Procedure verification.	17
8.	**NOTES**	**18**
8.1	Intended use.	18

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Check the source to verify that this is the current version before use.

CONTENTS

Paragraph Page

8.2 Subject term (key word) listing. 18
8.3 Changes from previous issue. 18
APPENDIX A Sample Charter for Nondestructive Inspection Requirements Review Board (NDIRRB) 19

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1. SCOPE

1.1 Scope.

This handbook provides guidance to government personnel and contractors for defining the requirements for establishing an acceptable Nondestructive Inspection (NDI) program for the development, production, and sustainment phases of DoD aircraft and missile structural components when directly referenced in the item specification contract or order. This handbook is for guidance only and cannot be cited as a requirement.

1.2 Applicability.

This handbook applies to structures for aircraft and missiles when the design activity or system specification requires nondestructive inspection for production acceptance, and recurring nondestructive inspection for long-term sustainment. This handbook may also be applicable to mechanical equipment, subsystems and propulsion systems but the guidance should be tailored for such use. This handbook applies to all phases of the system life cycle; acquisition, modification and sustainment. The NDI Program Plan should be developed at the beginning of the development phase and should define all NDI requirements to be adhered to during system development, test, production, and sustainment.

2. APPLICABLE DOCUMENTS

2.1 General.

The documents listed below are not necessarily all of the documents referenced herein, but are those needed to understand the information provided by this handbook.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein.

DEPARTMENT OF DEFENSE SPECIFICATIONS

JSSG 2006	Aircraft Structures
MIL-DTL-87929	Technical Manuals, Operation and Maintenance Instructions in Work Package Format (for USAF Equipment)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-867	Temper Etch Inspection
MIL-STD-1530	Aircraft Structural Integrity Program (ASIP)
MIL-STD-1537	Electrical Conductivity Test for Verification of Heat Treatment of Aluminum Alloys, Eddy Current Method
MIL-STD-1798	Mechanical Equipment and Subsystems Integrity Program
MIL-STD-1907	Inspection, Liquid Penetrant and Magnetic Particle, Soundness Requirements for Materials, Parts, and Weldments
MIL-STD-3024	Propulsion System Integrity Program (PSIP)

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DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-516	Airworthiness Certification Criteria
MIL-HDBK-1783	Engine Structural Integrity Program (ENSIP)
MIL-HDBK-1823	Nondestructive Evaluation System Reliability Assessment

(Copies of these documents are available online at https://assist.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia PA 19111-5094.)

2.2.2 Other Government documents.

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein.

AIR FORCE - ARMY – NAVY TECHNICAL MANUAL

T.O. 33B-1-2/ TM 1-1500-366-23/ NAVAIR 01-1A-16-2	Nondestructive Inspection General Procedures and Process Controls

(Copies of this document are available from the AF NDI Office, AFRL/RXSST, 4750 Staff Dr., Tinker AFB OK 73145, DSN 339-4931 or 405-739-4931.)

2.3 Non-Government publications.

The following documents form a part of this document to the extent specified herein.

ASTM INTERNATIONAL

ASTM B244	Standard Test Method for Measurement of Thickness of Anodic Coatings on Aluminum and of other Nonconductive Coatings on Nonmagnetic Basis Metals with Eddy-Current Instruments (DoD Adopted)
ASTM E164	Standard Practice for Contact Ultrasonic Testing of Weldments (DoD Adopted)
ASTM E213	Standard Practice for Ultrasonic Testing of Metal Pipe and Tubing
ASTM E215	Standard Practice for Standardizing Equipment for Electromagnetic Testing of Seamless Aluminum-Alloy Tube (DoD Adopted)
ASTM E309	Standard Practice for Eddy-Current Examination of Steel Tubular Products Using Magnetic Saturation (DoD Adopted)
ASTM E376	Standard Practice for Measuring Coating Thickness by Magnetic Field or Eddy-Current (Electromagnetic) Testing Methods (DoD Adopted)
ASTM E426	Standard Practice for Electromagnetic (Eddy-Current) Examination of Seamless and Welded Tubular Products, Austenitic Stainless Steel and Similar Alloys (DoD Adopted)
ASTM E427	Standard Practice for Testing for Leaks Using the Halogen Leak Detector (Alkali-Ion Diode)
ASTM	Standard Practice for Leaks Using the Mass Spectrometer Leak

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E498/E498M	Detector or Residual Gas Analyzer in the Tracer Probe Mode (DoD Adopted)
ASTM E1004	Standard Test Method for Determining Electrical Conductivity Using the Electromagnetic (Eddy-Current) Method (DoD Adopted)
ASTM E1417	Standard Practice for Liquid Penetrant Testing (DoD Adopted)
ASTM E1444/E1444M	Standard Practice for Magnetic Particle Testing (DoD Adopted)
ASTM E1742/E1742M	Standard Practice for Radiographic Examination (DoD Adopted)
ASTM E2033	Standard Practice for Computed Radiology (Photostimulable Luminescence Method)
ASTM E2580	Standard Practice for Ultrasonic Testing of Flat Panel Composites and Sandwich Core Materials Used in Aerospace Applications (DoD Adopted)
ASTM E2581	Standard Practice for Shearography of Polymer Matrix Composites, Sandwich Core Materials and Filament-Wound Pressure Vessels in Aerospace Applications (DoD Adopted)
ASTM E2582	Standard Practice for Infrared Flash Thermography of Composite Panels and Repair Patches Used in Aerospace Applications (DoD Adopted)
ASTM E2662	Standard Practice for Radiologic Examination of Flat Panel Composites and Sandwich Core Materials Used in Aerospace Applications (DoD Adopted)
ASTM E2698	Standard Practice for Radiological Examination Using Digital Detector Arrays
ASTM E2736	Standard Guide for Digital Detector Array Radiology
ASTM E2737	Standard Practice for Digital Detector Array Performance Evaluation and Long-Term Stability

(Copies of these documents are available from www.astm.org or ASTM International 100 Barr Harbor Dr., West Conshohocken, PA 19428-2959.)

AEROSPACE INDUSTRIES ASSOCIATION

AIA/NAS 410	NAS Certification and Qualification of Nondestructive Test Personnel (DoD Adopted)

(Copies of this document are available from www.aia-aerospace.org or Aerospace Industries Association, 1000 Wilson Blvd, Suite 1700, Arlington VA 22209-3928.)

SAE INTERNATIONAL

AMS2647	Fluorescent Penetrant Inspection Aircraft and Engine Component Maintenance
AMS-I-83387	Inspection Process, Magnetic Rubber (DoD Adopted)

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AMS-STD-2154 Inspection, Ultrasonic, Wrought Metals, Process For (DoD Adopted)

(Copies of these documents are available from www.sae.org or SAE World Headquarters, 400 Commonwealth Dr., Warrendale PA 15096-0001.)

3. DEFINITIONS

3.1 Inspector certification.

A written statement from the certifying agency verifying that an individual inspector has met applicable requirements. The certifying agency may be the employer of the inspection personnel.

3.2 Classification.

Classification refers to functional reliability requirements of the part and implies a confidence level requirement for NDI. A high-reliability class may require redundant testing to assure adequate NDI confidence level. Complex components may be divided into zones with separate classifications.

3.3 Contracting agency.

A contractor, subcontractor or Government agency procuring product or services.

3.4 Contractor.

Organization having contractual responsibility to the Government.

3.5 Final inspection.

The last inspection of a part or component usually, just prior to final acceptance. This inspection may occur during manufacturing or depot maintenance if the component will become uninspectable at some later stage in fabrication or assembly and is not subject to reinspection, or if it is inspected just after some processing step and is not subject to reinspection by the manufacturer after further processing.

3.6 Fracture-critical part.

A fracture-critical part is a safety-of-flight, structural component that is not single load path nor sized by durability or damage tolerance requirements but requires special emphasis due to criticality of the component (reference MIL-STD-1530, MIL-STD-3024, MIL-HDBK-516 and JSSG 2006).

3.7 Fracture-critical traceable part.

A fracture-critical traceable part is a safety-of-flight structural component that is either single-load path or sized by durability or damage tolerance requirements.

3.8 Maintenance-critical part.

A maintenance-critical part is a structural component whose failure will not cause a safety-of-flight condition but is sized by durability requirements and would not be economical to repair or replace.

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3.9 Mission-critical part.

A mission-critical part is a structural component in which damage or failure could result in the inability to meet critical mission requirements or could result in a significant increase in vulnerability.

3.10 Material Review Board (MRB).

A government approved review of discrepant materials by authorized contractor Engineering and Quality Assurance personnel to determine whether material or component can be returned to an acceptable state.

3.11 Nondestructive inspection (NDI).

NDI is the inspection of a structure or component in any manner that will not impair its future usefulness. The purpose of NDI may be to detect flaws at or beneath the external surface of a part, measure geometric characteristics, determine material structure or composition, or it may characterize physical, electrical, or thermal properties without causing changes in the part. The five primary methods of NDI are liquid penetrant, magnetic particle, eddy current, ultrasonic and radiography. Other methods include but are not limited to shearography and thermography.

3.12 Nondestructive inspection (NDI) procedure.

NDI procedures are contained in specifications that provide detailed information on "how-to" perform a particular NDI method (see 5.3.4).

3.12.1 General NDI procedure.

General NDI procedures are non-part specific nondestructive inspection procedures applicable to common product forms such as plates, composite laminates and bonded assemblies, bar stock, fasteners and tubular products, etc. General NDI procedures may also include procedures that describe standardization and inspection processes for common inspection methods.

3.12.2 Part specific NDI procedures.

Specific NDI procedures are procedures for inspection of a part or component, inspection of designated components, which incorporates all processing criteria and may be used in lieu of company process specifications.

3.13 NDI facility.

NDI facilities are organizations responsible to the System Program Offices (SPOs), contractor, and/or the subcontractor for nondestructive inspection services.

3.14 NDI Program Plan.

Document that defines the approach for implementation of nondestructive inspection processes during design, engineering development, production, and sustainment (i.e., in-service operation).

3.15 NDI process specification.

NDI process specifications define the nondestructive testing requirements for equipment, reference standards, materials, personnel, etc., for a particular NDI method or for the testing of a particular product.

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3.16 NDI Requirements Review Board (NDIRRB).

The Nondestructive Inspection Requirements Review Board (NDIRRB) is comprised of NDI, durability and damage tolerance, design, materials and processes (M&P) subject matter experts (SMEs) representing the prime, major subcontractors and government/customer. The NDIRRB should be formed early in the conceptual phase of a weapons system to provide guidance to the prime contractor, major sub-contractors, System Program Office (SPO), and subsequently to the System Program Manager (SPM), on all technical matters necessary to establish, implement and maintain an effective life cycle NDI Program responsive to the quality assurance, structural integrity and sustainment requirements of the program. The NDIRRB should review and assess inspection requirements and processes during all phases of the program from initial concept development through production and in-service operation.

3.17 Probability of detection (POD).

Probability of detection is a statistical measurement of the likelihood, with a specified confidence level, of finding a flaw of a defined size using a specific inspection process.

3.18 Qualification.

Qualification is the verified ability of personnel or products to meet the minimum capability, technical or performance requirements for a specified level of capability.

3.19 Safety-of-flight structure.

Safety-of-flight structures are structures whose single failure could cause loss of the weapon system, aircrew, or cause inadvertent store release. The loss could occur either immediately upon failure or subsequently if the failure remained undetected.

3.20 Subcontractors.

Subcontractors are organizations responsible to the contractor for a portion of the overall government contracted product.

3.21 Supplier.

Suppliers are organizations directly responsible for delivering a material, part or service to the Government, a contractor, or a subcontractor.

3.22 Validation.

The process of demonstrating the applicability of a proposed inspection process (equipment and procedures) to meet a specified requirement.

3.23 Verification.

The process of certifying the efficacy of an inspection process (equipment, reference standards and written procedures) through a demonstration, on a representative structure, in a representative environment, and by representative inspection personnel.

4. GENERAL REQUIREMENTS

The acquisition or modification of aircraft or missiles and their propulsion system should stress the inclusion and use of Nondestructive Inspection (NDI) in major design packages in production, test and sustainment environments. A detailed life cycle Nondestructive Inspection plan should be included as an integral component of the structural maintenance plans required by MIL-STD-1530.

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4.1 Preparation of the NDI Program Plan.

The prime contractor should establish in writing an overall systems plan to assure adequate nondestructive inspection of structural components for which the prime contractor determines NDI acceptance is necessary in an aircraft, missiles and their propulsion systems. This systems plan should include inspections throughout all phases of the program (design, engineering development, production and sustainment). The NDI Program Plan should be subject to the review by the NDIRRB and approval by the Government.

4.2 Responsibility for NDI.

Unless otherwise specified in the contract, the contractor is responsible for the performance of all production inspection requirements as specified in the NDI Program Plan. Except as otherwise specified in the contract, the contractor may use their own or any other facilities suitable for the performance of the inspection requirements unless disapproved by the Government. The contractor will be responsible for NDI sustainment tasks as specified in the contract.

The NDI Program should consider and implement appropriate nondestructive inspection processes into all phases of the program (design, engineering development, production, and sustainment). The program should establish a Nondestructive Inspection Requirements Review Board (NDIRRB) responsible for oversight and execution of the program.

4.2.1 Objective.

The objective of the NDI Program Plan is to assure a level of nondestructive inspection capability for both production and sustainment inspections, consistent with the prime contractor's design requirements.

4.2.2 Applicability.

The NDI program plan should include and be applicable to structural components as specified in the system specification and produced by the contractor, subcontractors and suppliers under government contract.

4.2.3 Elements.

The NDI Program Plan should present the scheme for establishing the NDI requirements and implementing procedures to meet these requirements. It should include the means of:

a. Conducting parts classification.

b. Coordination of design requirements and production NDI procedures.

c. Preparation and approval of production NDI procedures.

d. Demonstration of production NDI capability.

e. Qualification of NDI vendors to meet production capability requirements.

f. Obtain and maintain qualified NDI staff and an accredited NDI training/certification program.

g. Implementation of production NDI procedures.

h. Coordination and documentation of inspection requirements to support long-term sustainment including structural repairs.

i. Preparation of NDI technical data for long term sustainment.

j. Validation and verification of inspection technical data for sustainment.

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k. Assessment and documentation of sustainment NDI capability.

l. Development of NDI methods and procedures to support test articles.

m. Qualification of new technologies for use during production or component/aircraft testing or during the service life of the system.

n. Qualification of NDI methods adapted for the purpose of insitu structural damage sensing.

4.2.4 Coordination.

This plan should be coordinated with the contractor and customer Aircraft Structural Integrity Plan (ASIP) managers and the NDI Requirements Review Board (NDIRRB) when MIL-STD-1530 is a contractual requirement or with the contractor, customer Propulsion Integrity Plan (PSIP) managers and NDIRRB when MIL-STD-3024 is invoked.

4.3 Parts classification.

4.3.1 Structural parts classification.

The contractor should classify all structural components according to the structural integrity requirements of MIL-STD-1530. Complex components may be divided into zones and a separate classification or quality grade assigned to each zone in accordance with the reliability requirements. Classification should be noted on the drawing of the component or other released engineering data. Components should be classified as follows:

Fracture-Critical	A safety-of-flight structural component that is not single load path nor sized by durability or damage tolerance requirements but requires special emphasis due to the criticality of the component.
Fracture-Critical Traceable	A safety-of-flight structural component that is either single load-path or sized by durability or damage tolerance requirements the single failure of which would result in the loss of an aircraft or missile system.
Mission-Critical	A structural component in which damage or failure could result in the inability to meet critical mission requirements or could result in a significant increase in vulnerability.
Maintenance-Critical	A structural component whose failure will not cause a safety-of-flight condition but is sized by durability requirements and would not be economical to repair or replace.

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4.3.2 Propulsion systems parts classification.

When this handbook is used to establish an NDI Program plan for propulsion system components the parts classification requirements of MIL-STD-3024 should be used.

4.3.3 Mechanical equipment and subsystem parts classification.

When this handbook is used to establish an NDI Program plan for mechanical equipment or subsystem components, the parts classification requirements of MIL-STD-1798 should be used.

4.4 NDI Requirements Review Board (NDIRRB).

The contractor should establish a Nondestructive Inspection Requirements Review Board (NDIRRB) for oversight and execution of the NDI Program. The NDIRRB should be formed early in the design phase to review and assess product form concepts for inspectability in terms of process control and quality monitoring. The NDIRRB should also be responsible for review and approval of inspection methods and detectability assumptions implemented in the Force Structures Maintenance Plan. The board's decisions are subject to governmental/customer approval. The government/customer should retain the right of disapproval of NDIRRB decisions.

4.4.1 Purpose.

The NDIRRB should:

a. Review and approve the NDI Program Plan

b. Production Inspections: Determine the nondestructive inspection requirements for those components identified in accordance with 4.3 and assure that the most appropriate inspection methods(s) have been selected for the components being tested, and that the level of inspection is commensurate with the quality required.

c. Sustainment Inspections: Determine the nondestructive inspection processes, procedures and technologies required to maintain the integrity of components throughout their service life.

d. Determine the requirements for demonstration of NDI capability (e.g., MIL-HDBK-1823) for inspection processes implemented in all phases of the program.

4.4.2 NDI Requirements Review Board (NDIRRB) Membership.

The NDIRRB should be comprised of prime contractor, major subcontractor and government NDI, ASIP design and materials and processes (M&P) experts. The government program office should designate an independent government team to participate as representatives to the NDIRRB. The government representatives should also include subject matter experts (SMEs) from the respective service's NDI Program Office, research laboratories, depot maintenance facilities, and using commands. The NDIRRB may be augmented with independent non-governmental SMEs as required. The mission of this board is advisory and thus has no authority to direct a contractor, subcontractor, or government agency to take any action as a result of its findings. The NDIRRB provides recommendations to the prime contractor, major subcontractors, SPO and/or SPM.

5. DETAILED REQUIREMENTS FOR PRODUCTION

5.1 Documentation of the review.

Notation indicating NDIRRB review and approval should be documented on the appropriate drawing for each fracture-critical, fracture-critical traceable and mission critical structural

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component (see 4.3). Acceptable defect size, critical location and orientations should be referenced on the drawings or other released engineering specifications as appropriate. In all cases, the notation should consist of the signature of the contractor's authorized NDI Level III representative to the NDI Requirements Review Board.

5.2 Drawings.

The NDIRRB review should be the basis for specifying NDI requirements on engineering production drawings. The NDI procedure number, process specification, or coding, thereto, will be referenced on the drawing.

5.3 Preparation of NDI process and procedures specifications.

5.3.1 Use of general NDI process specifications.

The use of process specifications, such as those listed in 2.2 and 2.3, as sole controlling specifications is not permitted. These specifications reflect minimum quality requirements and are broad in scope.

5.3.2 Company NDI process specifications.

Company process specifications should be prepared incorporating the requirements of the referenced Government and non-Government process specifications. The company process specifications should meet or exceed the requirements of the referenced process specifications using the particular equipment, personnel and test facilities required to meet the reliability requirements of the product. If no general process specification exists for a particular method a company process specification or general NDI procedure should be generated and should incorporate sufficient information and criteria to adequately describe the NDI method and control the process.

5.3.2.1 Special NDI procedures.

Special procedures to inspect designated components may be used to supplement company process specifications. A contractor may elect to incorporate all processing criteria into each NDI procedure in lieu of generating process specifications.

5.3.2.2 NDI standardization.

The company process specification should reflect procedures, acceptance criteria, and records requirements to assure adequate quality assurance measures are being enforced to keep the NDI process in control. Basic processes, equipment, reference standards, materials, and method specific variables, as applicable, should be monitored to assure adequate control of the inspection process. Attempts should be made to use standard DoD NDI equipment and procedures (e.g., T.O. 33B-1-2/TM 1-1500-366-23/NAVAIR 01-1A-16-2) wherever possible to streamline the development and transition of in-service inspection procedures for sustainment.

5.3.2.3 Approval.

Company process specifications for NDI to be applied on aircraft and missile system components will be coordinated with an authorized representative of the contractor and be subject to review and approval by the NDIRRB and the Government as specified by the contract.

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5.3.3 NDI processes.

Company process specifications or other industry accepted processes may be used if approved by the NDIRRB. The following nondestructive inspection methods and the associated specifications are acceptable:

a. Magnetic particle, in accordance with ASTM E1444/E1444M, MIL-STD-1530 and MIL-STD-1907, as applicable.

b. Penetrant, in accordance with ASTM E1417, MIL-STD-1530, MIL-STD-1907, SAE AMS 2644, and SAE AMS 2647, as applicable.

c. Film Radiography, in accordance with ASTM E1742/E1742M.

d. Digital or Computed Radiography, in accordance with ASTM E2033, E2662, E2698, E2736, or E2737, as applicable.

e. Ultrasonic, in accordance with SAE AMS-STD-2154 and ASTM E213, E164 or E2580, as applicable.

f. Eddy current, in accordance with MIL-STD-1537 and ASTM B244, E215, E309, E376, E426 or E1004, as applicable.

g. Thermography, in accordance with ASTM E2582.

h. Magnetic rubber, in accordance with SAE-AMS-I-83387.

i. Leak testing, in accordance with ASTM E498 and ASTM E427, as applicable.

j. Temper etch, inspection in accordance with MIL-STD-867.

k. Shearography, in accordance with ASTM E2581.

5.3.4 NDI procedures.

Procedures will be provided for inspection of each part requiring NDI, except when special procedures are utilized. These procedures should be in accordance with the requirements of the component drawing, the company process specification or other engineering requirements and should contain the information listed below:

a. Specific part or drawing reference.

b. Specific part material; surface finish and part preparation, as applicable.

c. Manufacturer and model number of all instrumentation to be used, indicating optional equivalents, if desired.

d. Fixturing requirements, as applicable.

e. Manufacturer and identification of required inspection materials.

f. Reference to company process specification procedure if applicable.

g. Calibration/standardization procedure and reference standard identification as applicable.

h. Identification of areas to be inspected, detailed steps and level of inspector qualification required, and acceptable defect criteria including location and critical orientation (based upon primary stress condition and direction) or equivalent drawing or specification reference for these criteria.

i. Identifiable precautions in use of the inspection procedure.

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5.3.5 General NDI procedures.

General procedures or company process documents are acceptable for common product forms such as metallic plate, bar stock, fasteners, tubular products, as well as composite and hybrid material product forms. Contents of the NDI procedures may be modified as appropriate for general NDI procedures.

5.4 Implementation of NDI procedures.

5.4.1 Personnel.

The contractor should have available records of certification for personnel conducting and interpreting nondestructive inspections in accordance with the applicable sections of NAS 410 or other methods in accordance with the contract.

5.4.2 NDI reports.

The contractor's NDI reports and data records should be kept on file for the minimum amount of time as required by the contract. Reports should be signed or stamp identified by an authorized representative of the inspection facility. NDI techniques that produce digital records should be utilized on fracture critical and fracture critical traceable parts and assemblies where technically feasible. Digital NDI records (e.g., ultrasonic C-scans, radiographs, shearography images, etc.) used for final acceptance of fracture critical and fracture critical traceable parts (including assembly bond lines) should be delivered with each aircraft and be in a compatible electronic format as specified by the contract. Digital NDI records for fracture critical traceable parts should be traceable to aircraft number, part/assembly number, and part serial number, as applicable. Records should include NDI of repairs performed to achieve part conformance to requirements.

5.4.3 Equipment and materials.

The equipment and materials used for inspection should be in accordance with the applicable company process specification. Specified equipment should have adequate maintenance to assure that it is fully operational when used to conduct inspections.

5.4.4 NDI procedure verification.

All procedures and procedure families should be verified to assure repeatable defect detectability sufficient for the classification of the part. Procedures should be verified on parts or on test pieces simulating the actual part and which provide the essential features of the part with regard to the important application variables which may affect defect sensitivity and confidence level. Redundant inspections may be utilized.

5.4.5 Removal of discontinuities.

When nondestructive inspection reveals discontinuities in excess of the level permitted by applicable drawings or documents, such discontinuities may be removed if permitted by applicable drawings and documents or authorized by Materials Review Board action. Evidence of removal should be shown by reinspection.

5.4.5.1 Reinspection.

Reinspection for removal of discontinuities should be conducted using the same procedure. If a modified or different procedure is necessary, an addendum or temporary change to the original procedure action should be prepared and approved by the responsible NDI Level III and Materials Review Board showing the essential features of the repair test.

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5.4.6 Inspection stages.

5.4.6.1 First article inspections.

Inspections of all first article components should be conducted as part of manufacturing qualification and to insure the components meets the applicable engineering requirements. The requirements for first article inspections should be defined and may exceed requirements for subsequent receiving inspections.

5.4.6.2 Receiving inspections.

Incoming materials, parts or assemblies should be inspected to ensure they meet the applicable engineering requirements.

5.4.6.3 Manufacturing and assembly.

Inspections should be performed as necessary during manufacture and assembly of components to assure the quality of final parts or assemblies commensurate with the part or assembly classification and the criticality of the manufacturing processes. These inspections should be approved by the NDIRRB.

a. NDI should be performed subsequent to operations which may adversely affect the quality of material or part, such as heat treating, forging and cold working, When processing operations are involved which may interfere with the required inspection, inspections should be performed prior to such operations.

b. NDI should be performed when composite parts may be damaged as a result of machining operations or final assembly.

5.4.7 Data and documentation.

Requirements expressed or implied herein concerning preparation, submittal, approval, availability, retention, or delivery of data or documentation should be applicable only to the extent provided in the contract.

5.4.8 Vendor qualification.

All facilities, including sub-tier suppliers, performing inspection of fracture critical and fracture critical traceable structural components, must be qualified by the contractor to ensure their NDI processes and procedures meet design requirements. The NDI Program Plan should address the criteria for qualification, including facility and personnel capability demonstrations, required for each inspection method employed. The vendor's NDI process specifications and fracture-critical, fracture-critical traceable and mission-critical part specific procedures should be reviewed and approved by the contractor's Level III prior to the initial vendor qualification audit. The qualification process should include initial and periodic audits of the vendor's facility. The audits should be performed by personnel knowledgeable in the NDI methods being qualified, using a Level III approved checklist as a framework for conducting the audit. The auditor should witness the vendor performing sample fracture-critical, fracture-critical traceable and mission-critical part specific procedures as part of the audit. The initial audits of vendors manufacturing fracture-critical, fracture-critical traceable and mission-critical parts should be performed by an NDI Level III in that method. The recurring audits frequency may be dependent on past performance and audit results, but should not exceed two years for vendors of fracture-critical, fracture-critical traceable and mission-critical parts.

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5.4.9 Capability demonstration.

The capability of nondestructive inspection processes used for production process monitoring and quality control of components should be established, as required by MIL-STD-1530, to mitigate risk of missing defects. Special emphasis should be given inspection processes applied to fracture-critical, fracture-critical traceable and mission-critical parts. If a procedure is required to detect flaws smaller than the accepted capability for the particular method, a capability demonstration should be performed to prove that the technique can reliably detect flaws of that size. Capability demonstrations should also be required for qualification of new inspection processes or technologies prior to implementation. The requirement and approach for conducting capability studies should be addressed within the NDI Program Plan.

6. DETAILED REQUIREMENTS FOR TEST ARTICLES

6.1 First article testing.

Structural components undergoing first article testing should receive, as a minimum, production representative NDI. Additional NDI may be utilized to further quantify the component as part of the first article evaluation.

6.2 Static test articles.

Static test articles will receive, as a minimum, production representative NDI. Additional NDI may be utilized to further quantify the component post-test.

6.3 Component, full-scale durability test articles.

Structural components undergoing full-scale durability test should receive a production representative NDI prior to test. NDI should be utilized during component and full-scale durability testing. These tests provide a valuable opportunity to verify NDI procedures that may be required to sustain the aircraft. Additionally, the application of NDI during durability tests provides early detection of impending failures which facilitates more accurate determination of the onset of cracking the opportunity to monitor crack growth rates, the development and demonstration of repair concepts and the extension of the test by preventing premature catastrophic failure of the test article. Durability test articles provide an excellent opportunity to evaluate standard and emerging NDI technologies and embedded structural damage sensing (structural health monitoring) systems for use as sustainment tools once the system is fielded.

6.4 Teardown inspections.

NDI should be performed on component and full-scale test durability articles during the teardown inspections that are performed after cyclic testing.

6.5 Flight testing.

NDI should be utilized as required during flight testing to monitor fracture-critical, fracture-critical traceable and mission critical structures where durability and damage tolerance analysis identifies a shortfall. These tests also provide a valuable opportunity to verify NDI procedures that may be required to sustain the aircraft. The NDI procedures that support flight test should be developed considering their eventual inclusion into the sustainment requirements for the system. Inspection procedures developed to support flight test should reference the equipment available to the flight test maintenance organization and be written in accordance with 7.3.1.

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6.6 Test support monitoring inspections.

NDI should be utilized as necessary to assess the onset of damage other than fatigue, including corrosion, thermal and impact damage to components and structures. The test support NDI procedures should also be developed considering their eventual inclusion in the sustainment requirements for the system.

7. DETAILED REQUIREMENTS FOR SUSTAINMENT

7.1 Defining sustainment inspection requirements.

The initial and recurring inspections should be identified and inspection solutions should be implemented to maintain the safety of the structure throughout the intended design life. These inspections should be established in accordance with MIL-STD-1530, MIL-STD-3024 or MIL-STD-1789. Sustainment requirements should be considered throughout the design, development and pre-production phases. Inspection requirements for component, full-scale and flight test articles should be leveraged as opportunities to develop and validate sustainment inspection procedures.

7.2 Sustainment inspection requirements review.

The NDIRRB will:

a. Review the requirements and proposed inspection processes for initial and recurring in-service inspections and determine whether the proposed methods and the associated detection capabilities are sufficient to address the inspection requirements.

b. Review plans for conducting probability of detection (PoD) and/or proficiency studies to quantify detection capability used to establish realistic reinspection intervals.

c. Review and approve the capability assumptions used to establish inspection intervals for recurring inspections.

d. Recommend technology development or demonstration programs to identify and transition inspection solutions as required to address capability shortfalls.

e. Provide recommendations and guidance directly to the prime contractor, major subcontractors and SPO/SPM for decision making regarding all aspects of the NDI Program.

7.3 Preparation of NDI procedures.

The NDI Program will consider and implement appropriate nondestructive inspection processes to address long term sustainment requirements. Inspection procedures should be written to utilize existing standard-issue NDI methods and equipment to the greatest extent possible. Any proposed use of non-standard NDI methods/equipment should be technically justified to the Government including evidence to document its capability to meet the specific inspection requirements.

7.3.1 Sustainment NDI procedure content.

NDI procedures will be provided for inspection of each part requiring inspection. Each procedure should contain the following information:

a. Inspection purpose or intent.

b. Specific part or drawing reference.

c. Applicable weapon system configuration.

d. Specific part material; surface finish and part preparation, as applicable.

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e. Inspector training and qualification requirements.

f. Manufacturer and model number of all equipment to be used, indicating optional equivalents, if desired.

g. Fixturing requirements, as applicable.

h. Manufacturer and identification of required inspection materials.

i. Reference to inspection standard practices as applicable.

j. Calibration/standardization procedure, including initial equipment settings, and reference standard identification, including drawings, as applicable.

k. Inspection access including overall system figures leading to detailed access figures.

l. Inspection frequency (at what flight hours to initiate – and frequency thereafter).

m. Identification of expected damage type (fatigue crack, SCC, impact damage, corrosion, etc), location and orientation.

n. Identification of required scan direction and coverage.

o. Evaluation criteria.

p. Acceptance criteria (including repair criteria where applicable).

q. Component disposition.

r. Cautions and warnings in use of the inspection procedure.

s. Part marking (to designate/document if part has passed or failed inspection).

t. Post-test requirements (including cleaning, preservation, demagnetization, etc.).

7.3.2 Sustainment NDI procedure format.

NDI procedures will be written in accordance with MIL-DTL-87929 to the appropriate inspector training level or as stipulated by the contract.

7.4 Inspection capability demonstration.

The capability of nondestructive inspection processes to be used for long term sustainment should be established. Special emphasis should be given to fracture or fatigue and mission-critical parts. Capability demonstrations should also be performed when new materials, fabrications techniques, specialty coatings and repairs processes are implemented that may affect the efficacy of the selected inspection processes. The methodology for establishing capability should be addressed within the NDI Program Plan. Capability assumptions used to establish reoccurring inspection intervals should be approved by the NDIRRB and documented in accordance with the MIL-STD-1530, MIL-STD-3024 or MIL-STD-1789. Use of pre-established capability assumptions for established inspection methodologies (i.e., Structures Bulletin EN-SB-008-012) should be approved by the NDIRRB.

7.5 NDI procedure validation and verification.

The NDI Program Plan should address the process for validation and verification of all inspection procedures. All procedures and procedure families should be verified to assure repeatable damage detection capability sufficient for the classification of the part. Procedures may be verified on parts or on test pieces simulating the actual part and which provide the essential features of the part with regard to the important application variables which may affect defect detection sensitivity, signal-to-noise and confidence level. Use of substitute test pieces

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should be approved by the NDIRRB. If necessary, a POD study should be conducted to verify the detection capability of the NDI processes. Further guidance on POD studies can be found in MIL-HDBK-1823.

7.5.1 Procedure validation.

Procedure validation should be the responsibility of the author of the procedure unless contracted by a government organization. The process for performing and documenting procedure validations should be approved by the customer and should accomplish the following objectives:

a. Collect and analyze supporting data.

b. Establish appropriate method, equipment and procedure parameters.

c. Establish requirements for unique reference standards to effectively control set-up variability.

d. Establish appropriate technician skill-level for reliable procedure performance.

e. Establish appropriate component removal and surface preparation requirements necessary for adequate access.

f. Design/manufacture test specimens containing actual or simulated discontinuities.

g. Design/manufacture prototype support equipment.

h. Produce a draft technical procedure.

i. Demonstrate procedure feasibility to responsible NDI Level III (reference NAS 410).

j. Approval of the procedure by the responsible NDI Level III (reference NAS 410).

7.5.2 Procedure verification.

Procedure verification will be performed immediately following the conclusion of successful procedure validation. The verification process must prove the suitability of an NDI procedure to perform its intended purpose to the satisfaction of the end user, the responsible NDI Level III, the depot NDI program manager and the responsible program engineer. Verification will be performed on an aircraft or on an individual assembly or component as directed by the inspection procedure. Verification will also be conducted within a representative environment on actual aircraft parts or test pieces by representative inspectors.

Procedure verification must as a minimum accomplish the following objectives and be documented using a Qualification Checklist approved by the NDIRRB and government:

a. Ensure the procedure is thorough, understandable and logically written.

b. Ensure the procedure is understood and executable by the lowest skill level projected for flight test and in-service use.

c. Ensure the specified equipment performs as expected and is readily available to the intended end-user.

d. Ensure the specified part preparation procedures are logical and can be accomplished as written.

e. Demonstrate the procedure meets expectations for detection capability and false call propensity.

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f. Ensure the inspection results are properly documented and communicated to the appropriate engineering, maintenance, and other NDI functions.

g. Determine the need for specialized personnel training or certification.

h. Determine the estimated inspection process man-hours.

i. Account for human factor and environmental variables as necessary.

j. Document verification per the appropriate Qualification Checklist.

k. Approve the procedure for publication, distribution and use.

8. NOTES

8.1 Intended use.

This document is intended to be used by the applicable organization(s) in setting forth specific policy as to the required nondestructive inspection program to be conducted in the production and sustainment of aircraft, missiles and their propulsion systems.

8.2 Subject term (key word) listing.

Calibration

Eddy Current

Magnetic Particle

Penetrant

Radiographic

Shearography

Temper etch

Thermography

Ultrasonic

8.3 Changes from previous issue.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX A

SAMPLE CHARTER
FOR
NONDESTRUCTIVE INSPECTION REQUIREMENTS REVIEW BOARD (NDIRRB)

A.1 SCOPE

A.1.1 Scope.

This appendix provides the guidance for preparing a Charter for a system specific Nondestructive Inspection Requirements Review Board (NDIRRB). The following example charter defines the typical mandatory responsibilities and requirements of an NDIRRB as indicated by the verb “shall”. This text is provided as guidance only and should be tailored based on program contractual requirements.

The NDIRRB is comprised of government NDI, durability and damage tolerance, design, M&P SMEs representing the interests of the government/customer. The NDIRRB should be formed early in the conceptual phase of a weapons system to provide guidance to the System Program Office (SPO) and subsequently to the System Program Manager (SPM) on all technical matters necessary to establish, implement, and maintain an effective life cycle NDI Program responsive to the quality assurance, structural integrity and sustainment requirements of the program.

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APPENDIX A

CHARTER

FOR

(ENTER SYSTEM NAME)

NONDESTRUCTIVE INSPECTION REQUIREMENTS REVIEW BOARD (NDIRRB)

INTRODUCTION.

The appropriate application of Nondestructive Inspection (NDI) to aerospace systems helps achieve system safety and structural integrity requirements. To ensure appropriate NDI is implemented, a Nondestructive Requirements Review Board shall be established during the conceptual phase of each new major weapon system acquisition, major modifications to existing weapon systems, or as required as a means of achieving this goal. The intent is to bring together the designer, maintainer and user so that they may all contribute in formulating and implementing an effective, continuing system NDI plan. This charter establishes the NDIRRB for the (enter system name).

PURPOSE.

The NDIRRB shall be the authorized body for NDI policy and implementation. The purpose of the NDIRRB is to provide assistance and guidance to the System Program Office (SPO), System Program Manager (SPM), the Prime Contractor and Suppliers on all technical matters necessary to establish, implement, and maintain an effective NDI Program responsive to the structural integrity and in-service maintenance requirements of the (enter system name). Each NDIRRB team member is responsible to be cognizant of the NDI process at the Prime Contractor and its suppliers/subcontractors and bring any NDI program considerations/problems to the review board. This could require the Prime Contractor and suppliers/subcontractors to present NDI considerations/problems to the NDIRRB for resolution.

MEMBERSHIP.

The NDIRRB shall be comprised of prime contractor, major subcontractor and government NDI, ASIP, design and materials and processes (M&P) experts. The government program office shall designate an independent government team to participate as representatives to the NDIRRB. The government representatives shall also include subject matter experts (SMEs) from the respective service's NDI Program Office, research laboratories, depot maintenance facilities, and using commands. The NDIRRB may be augmented with independent non-governmental SMEs as required.

This NDIRRB is chaired by Prime Contractors Lead for Materials and Process Engineering or Lead NDI Level III. It is a prerequisite that each permanent member possess extensive background in NDI to maximize the efficiency of the board. In addition, representative from structural engineering, quality, material procurement, design, manufacturing and other organizations including the supplier's Responsible NDI Level III may be requested to participate. The NDIRRB shall consist of the following members and alternates from the organizations listed. The NDIRRB shall update the policy content and membership as required.

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MEMBER		ALTERNATE
(Enter name)	Prime Contractor - M&P Engineer/NDI Level III	(enter name/org)
(Enter name)	Prime Contractor - Product Quality Engineering	(enter name/org)
(Enter name)	Prime Contractor - Product Quality NDI Engineer	(enter name/org)
(Enter name)	Prime Contractor - Lead Test Engineer	(enter name/org)
(Enter name)	Prime Contractor - DADT Engineering	(enter name/org)
(Enter name)	Customer - NDI Subject Matter Experts	(enter name/org)
(Enter name)	Customer - DADT Engineering	(enter name/org)
(Enter name)	AF/Navy NDI Program Office	(enter name/org)
(Enter name)	User Command	(enter name/org)
(Enter name)	Subcontactor NDI Engineering (as applicable)	(enter name/org)
(Enter name)	Prime Contractor - Technical Data Manager	(enter name/org)

RESPONSIBILITIES.

The mission of this board is advisory and thus has no authority to direct a contractor, subcontractor, or government agency to take any action as a result of its findings. Rather, the Board shall make recommendations directly to the prime contractor, major subcontractors, SPO and/or after Program Management Responsibility Transfer (PMRT) directly to the SPM. The NDIRRB shall present its recommendations in written form coordinated through all board members. There shall be united agreement on all decisions and recommendations made by the NDIRRB. The responsibilities of the NDIRRB are summarized as follows:

a. Convene at regularly scheduled intervals throughout the life cycle of this system at the times and places arranged by the chairperson. The regularly scheduled intervals will be quarterly unless the chairperson, with a majority of the members, determines a different schedule is necessary or the quarterly meeting coincides with a trip to other scheduled meeting(s) or events(s). In any case, the NDIRRB shall meet an absolute minimum of once a year during conceptual phase and semiannually thereafter.

b. The NDIRRB shall advise the System Program Office, prime contractor and major subcontactors on the following:

(1) Adequacy of existing NDI technologies (methods and equipment) and the use of new technologies as they become available.

(2) Ensure that Nondestructive Inspection is considered throughout all phases of the weapon system life cycle.

(3) Program NDI is consistent with contractual requirements.

(4) Report status to the Program management and DoD NDI representatives on NDI issues when significant changes occur.

(5) Compliance of the NDI program with applicable standards, specifications, design handbooks, and related documentation.

(6) Evaluation of proposed life cycle NDI Plan during source selection.

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(7) Contractual NDI requirements during the conceptual refinement, technical development, system development and demonstration, and production or deployment phases of this system's acquisition.

(8) NDI or NDI-related data item descriptions (DIDs) (Examples of these are: Structural Integrity Plan, NDI Program Plan, Damage Tolerance/Durability Control Plan, Corrosion Program Plan and Force Maintenance Plan, etc.).

(9) Status of NDI Program and DIDs at Preliminary Design Review (PDR).

(10) Recommended action and NDI Program and DIDs at Critical Design Review (CDR), including NDI program assessment against maintenance NDI requirements (examples of these include: Initial flaw size and NDI capability assumptions used to establish initial and reoccurring inspections, etc.).

(11) Adequacy of contractor and subcontractor compliance with MIL-HDBK-6870 and the NDI Program Plan for raw material purchases, component fabrication, and system assembly during production/manufacturing.

(12) System-peculiar NDI technical order development and subsequent in-process reviews. Review coordination with related technical manuals. Participation in validation and verification of NDI technical orders.

(13) Training requirements and training manuals as they relate to system-peculiar NDI equipment and processes.

c. Conduct on-site reviews at contractor and subcontractor facilities to evaluate the use of NDI during manufacturing.

d. Conduct on-site reviews at flight test, climatic test, and operational sites to evaluate NDI discrepancies or problem areas and the impact on Air Force field or depot facilities.

e. Maintain a continuing record of all board meetings, trips and actions, to include but not limited to:

(1) Specific area(s) reviews.

(2) Technical discussion, including advantages and disadvantages of feasible alternative courses of action.

(3) Board recommendations and their cost/schedule impact.

(4) Individual or agency responsibility.

(5) Final disposition of board recommendations.

PROGRAM MANAGER RESPONSIBILITY TRANSFER (PMRT).

The NDIRRB shall review the life cycle NDI program data and NDI technical orders package to ensure they meet the requirements of the operating command and SPMs of the Air Logistics Center (ALC) or Depot gaining the new system. Recommended actions or changes shall be provided to the procuring activity before CDR. After PMRT, the ALC System Program Manager shall name the NDIRRB Chairperson and amend this Charter.

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CONCLUDING MATERIAL

Custodians:

Army – MR

Navy - AS

Air Force - 11

Preparing activity:

Air Force - 11

Project No. NDTI-2011-004

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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